Practical – 07

Aim: Program to implement LALR Parsing in C.

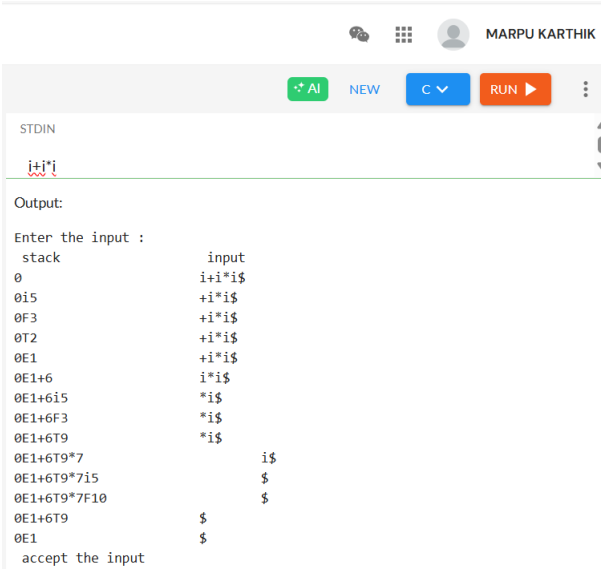
Program:

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>

void push(char *, int *, char);
char stacktop(char *);
void isproduct(char, char);
int ister(char);
int  isnter(char);
int isstate(char);
void error();
void isreduce(char, char);
char pop(char *, int *);
void printt(char *, int *, char [], int);
void rep(char [], int);

struct action {
   char row[6][5];
};

const struct action A[12] = {
   {"sf", "emp", "emp", "se", "emp", "emp"},
   {"emp", "sg", "emp", "emp", "emp", "acc"},
   {"emp", "rc", "sh", "emp", "rc", "rc"},
   {"emp", "re", "re", "emp", "re", "re"},
   {"sf", "emp", "emp", "se", "emp", "emp"},
   {"emp", "rg", "rg", "emp", "rg", "rg"},
   {"sf", "emp", "emp", "se", "emp", "emp"},
   {"sf", "emp", "emp", "se", "emp", "emp"},
   {"emp", "sg", "emp", "emp", "sl", "emp"},
   {"emp", "rb", "sh", "emp", "rb", "rb"},
   {"emp", "rb", "rd", "emp", "rd", "rd"},
   {"emp", "rf", "rf", "emp", "rf", "rf"}
};

struct gotol {
   char r[3][4];
};

const struct gotol G[12] = {
   {"b", "c", "d"},
   {"emp", "emp", "emp"},
   {"emp", "emp", "emp"},
```

```
    {"emp", "emp", "emp"},
    {"i", "c", "d"},

    {"emp", "emp", "emp"},
    {"emp", "j", "d"},
    {"emp", "emp", "k"},
    {"emp", "emp", "emp"},
    {"emp", "emp", "emp"},
};

char ter[6] = {'i', '+', '*', ')', '(', '$'};
char nter[3] = {'E', 'T', 'F'};
char states[12] = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'm', 'j', 'k', 'l'};
char stack[100];
int top = -1;
char temp[10];

struct grammar {
    char left;
    char right[5];
};

const struct grammar rl[6] = {
    {'E', "e+T"},
    {'E', "T"},
    {'T', "T*F"},
    {'T', "F"},
    {'F', "(E)"},
    {'F', "i"},
};

int main() {
    char inp[80], x, p, dl[80], y, bl = 'a';
    int i = 0, j, k, l, n, m, c, len;
    printf("Enter the input :");
    scanf("%s", inp);
    len = strlen(inp);
    inp[len] = '$';
    inp[len + 1] = '\0';
    push(stack, &top, bl);
    printf("\n stack \t\t\t input");
    printt(stack, &top, inp, i);

    do {
        x = inp[i];
        p = stacktop(stack);
        isproduct(x, p);
        if (strcmp(temp, "emp") == 0)
            error();
        if (strcmp(temp, "acc") == 0)
            break;
```

```
        else {
            if (temp[0] == 's') {
                push(stack, &top, inp[i]);

                push(stack, &top, temp[1]);
                i++;
            }
            else {
                if (temp[0] == 'r') {
                    j = isstate(temp[1]);
                    strcpy(temp, rl[j - 2].right);
                    dl[0] = rl[j - 2].left;
                    dl[1] = '\0';
                    n = strlen(temp);
                    for (k = 0; k < 2 * n; k++)
                        pop(stack, &top);
                    for (m = 0; dl[m] != '\0'; m++)
                        push(stack, &top, dl[m]);
                    l = top;
                    y = stack[l - 1];
                    isreduce(y, dl[0]);
                    for (m = 0; temp[m] != '\0'; m++)
                        push(stack, &top, temp[m]);
                }
            }
        }
        printt(stack, &top, inp, i);
    } while (inp[i] != '\0');

    if (strcmp(temp, "acc") == 0)
        printf(" \n accept the input ");
    else
        printf(" \n do not accept the input ");
    getchar();
}

void push(char *s, int *sp, char item) {
    if (*sp == 100)
        printf(" stack is full ");
    else {
        *sp = *sp + 1;
        s[*sp] = item;
    }
}

char stacktop(char *s) {
    if (top >= 0)
        return s[top];
    return '\0'; // Add a default return value
}
```

```
void isproduct(char x, char p) {
    int k, l;
    k = ister(x);
    l = isstate(p);

    strcpy(temp, A[l - 1].row[k - 1]);
}

int ister(char x) {
    int i;
    for (i = 0; i < 6; i++)
        if (x == ter[i])
          return i + 1;
    return 0;
}

int isnter(char x) {
    int i;
    for (i = 0; i < 3; i++)
        if (x == nter[i])
          return i + 1;
    return 0;
}

int isstate(char p) {
    int i;
    for (i = 0; i < 12; i++)
        if (p == states[i])
           return i + 1;
    return 0;
}

void error() {
    printf(" error in the input ");
    exit(0);
}

void isreduce(char x, char p) {
    int k, l;
    k = isstate(x);
    l = isnter(p);
    strcpy(temp, G[k - 1].r[l - 1]);
}

char pop(char *s, int *sp) {
    char item;
    if (*sp == -1)
        printf(" stack is empty ");
    else {
        item = s[*sp];
        *sp = *sp - 1;
```

```
    }
    return item;
}
void printt(char *t, int *p, char inp[], int i) {
    int r;
    printf("\n");
    for (r = 0; r <= *p; r++)
        rep(t, r);
    printf("\t\t\t");
    for (r = i; inp[r] != '\0'; r++)
        printf("%c", inp[r]);
}
void rep(char t[], int r) {
    char c;
    c = t[r];
    switch (c) {
        case 'a': printf("0"); break;
        case 'b': printf("1"); break;
        case 'c': printf("2"); break;
        case 'd': printf("3"); break;
        case 'e': printf("4"); break;
        case 'f': printf("5"); break;
        case 'g': printf("6"); break;
        case 'h': printf("7"); break;
        case 'm': printf("8"); break;
        case 'j': printf("9"); break;
        case 'k': printf("10"); break;
        case 'l': printf("11"); break;
        default: printf("%c", t[r]); break;
    }
}
```

OUTPUT: